Pocket Packet Sniffer

Software Test Plan

WiFi / Ethernet / BLE Portable Packet Sniffing & Dashboard Tool

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1. Test Plan Identifier

Field	Value
Document ID	PPS-TP-1.0
Related SRS	PPS-SRS-1.0
Related SADS	PPS-SADS-1.0
Standard Followed	IEEE Std 829-2008, Standard for Software and System Test Documentation
Project	Pocket Packet Sniffer (Mini-Project, Part 1)

2. Introduction

2.1 Objectives

This Test Plan defines the scope, approach, resources, and schedule of testing activities for the Pocket Packet Sniffer. Its objective is to verify that all functional requirements (FR-1 through FR-15) and non-functional/security requirements (NFR-*, SEC-REQ-*) defined in the SRS (PPS-SRS-1.0) are correctly and verifiably implemented before Part 1 sign-off, and to establish traceability between requirements, design, and test cases.

2.2 Background

The Pocket Packet Sniffer combines embedded capture hardware (Raspberry Pi + ESP32), the Zeek analysis engine, and a Python-based dashboard. Because the product captures and analyzes live network and radio traffic and stores potentially sensitive data, testing must cover functional correctness, real-time performance under load, and security controls with equal rigor.

2.3 Test Levels in Scope

- Unit Testing – individual Python modules (parser, rule engine, API handlers).
- Integration Testing – capture module → Zeek → backend → dashboard data flow.
- System Testing – end-to-end scenarios on the assembled hardware/software stack.
- Security Testing – authentication, encryption, RBAC, and audit-log validation (Section 5.1).
- Acceptance Testing – validation against SRS requirements prior to stakeholder sign-off.

3. Test Items

The following software and hardware items are within the scope of this test plan, at the version/build to be frozen for the Part 1 milestone:

Item	Description	Version/Build
Capture Firmware	ESP32 WiFi/BLE promiscuous-mode capture firmware	v0.1 (Part 1 build)
Backend Service	Python REST/WebSocket API, rule engine, alert manager	v0.1 (Part 1 build)
Zeek Integration Layer	Zeek script bundle + log ingestion adapter	v0.1 (Part 1 build)
Dashboard UI	Web-based single-page dashboard application	v0.1 (Part 1 build)
Storage & Auth Subsystem	SQLite schema, encryption module, RBAC/session manager	v0.1 (Part 1 build)
Reference Hardware Unit	Raspberry Pi 4B + ESP32 capture module + battery pack	Prototype Unit #1

4. Features to Be Tested / Not to Be Tested

4.1 Features to Be Tested

- WiFi, Ethernet, and BLE packet capture (FR-1, FR-2, FR-3, FR-15).
- Capture filter configuration (FR-4).
- Zeek-based traffic analysis and log generation (FR-5).
- Live dashboard statistics and refresh behavior (FR-6).
- Capture session start/stop control (FR-7).
- PCAP export and session report generation (FR-8, FR-9).
- Anomaly detection: ARP spoofing, port scan, rogue AP, BLE flood (FR-10).
- Security alert raising and acknowledgment (FR-11, FR-12).
- Storage/retention administration (FR-13).
- Offline local-AP connectivity (FR-14).
- Performance, reliability, usability, and maintainability NFRs (Section 5 of SRS).
- Authentication, RBAC, encryption at rest/in transit, audit logging, account lockout (SEC-REQ-1 through SEC-REQ-6).

4.2 Features Not to Be Tested (Part 1 Scope Exclusions)

- Multi-device fleet management / centralized cloud dashboard (deferred to a later phase, out of Part 1 scope).
- Firmware over-the-air update mechanism for the ESP32 module (planned for Part 2).
- Third-party SIEM export integrations (Splunk/Elastic connectors) – not part of Part 1 functional baseline.
- Long-term (multi-week) endurance/soak testing of storage wear – deferred to hardware qualification phase.

5. Approach (Test Strategy)

Testing follows a layered strategy proceeding from isolated unit tests through integration and system tests to a final acceptance pass mapped directly to SRS requirement IDs. Automated tests (pytest for backend/unit and integration; Selenium/Playwright for dashboard UI) are combined with manual exploratory and security testing where automation is impractical (e.g., RF capture range, physical device tampering). Every test case in Section 6 records the SRS requirement ID(s) it verifies, ensuring bidirectional traceability. Entry criteria for system testing require all unit and integration tests to pass on the target hardware build; exit criteria require 100% execution of the test cases in Section 6 with no open Severity-1 (Critical) or Severity-2 (High) defects.

5.1 Security Validation Approach

Security testing is treated as a first-class track rather than a subset of functional testing, given the sensitivity of captured traffic. The approach combines:

- Authentication & Session Testing – verify credential policy enforcement, session token expiry, and account lockout after 5 failed attempts (SEC-REQ-2, SEC-REQ-6).
- Access Control Testing – attempt privilege-escalation and horizontal-access scenarios to confirm RBAC boundaries between Field Engineer and Administrator roles (SEC-REQ-3).
- Cryptographic Validation – inspect stored .pcap files and the metadata database directly on disk to confirm AES-256 encryption at rest (SEC-REQ-1); capture and inspect dashboard/API traffic with a network analyzer to confirm TLS 1.2+ is enforced and no plaintext fallback exists (SEC-REQ-4).
- Audit Log Integrity Testing – attempt to modify or delete audit-log entries and confirm the hash-chain detects tampering (SEC-REQ-5).
- Vulnerability/Exposure Scanning – run automated scans (e.g., Nmap service enumeration, OWASP ZAP baseline scan against the dashboard API) to identify unintended open ports or common web vulnerabilities.
- Negative/Abuse-Case Testing – malformed API input, oversized capture filter strings, and replayed authentication tokens are submitted to confirm safe rejection rather than crash or bypass.

Security test cases are enumerated in Section 6 (TC-SEC-01 through TC-SEC-03) and are traced to SEC-OBJ-1/2/3 and SEC-REQ-1 through SEC-REQ-6 defined in the SRS.

6. Item Pass/Fail Criteria

- A test case passes only if the observed result exactly matches the documented expected result for every step.
- A feature is considered verified only when all of its associated test cases pass on the reference hardware build.
- The overall test cycle passes when ≥ 95% of all test cases pass, with zero open Critical (Sev-1) or High (Sev-2) defects and no unresolved security findings from Section 5.1.
- Any failed security test case (TC-SEC-*) is treated as Severity-1 by default regardless of apparent impact, pending triage.

7. Suspension Criteria and Resumption Requirements

- Testing is suspended if a Severity-1 defect blocks execution of more than 30% of remaining planned test cases (e.g., capture pipeline crash, authentication bypass).
- Testing is suspended if the reference hardware unit becomes physically unavailable or damaged.
- Testing resumes once the blocking defect is fixed, regression-tested by the developer, and a new build is deployed to the reference hardware.

8. Test Deliverables

- This Test Plan document (PPS-TP-1.0).
- Test case specifications and traceability matrix (Section 6).
- Test execution logs and defect reports (tracked per cycle).
- Security test evidence (scan reports, packet captures used for TLS/encryption verification).
- Final test summary report at the close of Part 1.

9. Testing Tasks

Task	Description	Owner
Test case design	Derive test cases from SRS requirement IDs; build traceability matrix	QA/Test Lead
Test environment setup	Configure reference hardware, isolated WiFi test range, traffic generators	Hardware/DevOps
Unit & integration test automation	Implement pytest suites for backend, rule engine, API	Backend Developers
UI automation	Implement dashboard smoke/regression tests (Playwright)	Frontend Developer
Security test execution	Execute Section 5.1 security validation activities	Security Tester
Defect triage & regression	Log, prioritize, and verify fixes for failed test cases	QA/Test Lead
Test summary reporting	Compile results, coverage, and open-issue status	QA/Test Lead

10. Environmental Needs

- Reference hardware unit: Raspberry Pi 4B, ESP32 capture module, Ethernet switch/cable, BLE beacon devices for test stimulus.
- Isolated test WiFi network (dedicated SSID) to avoid interference with production networks during capture and anomaly-injection tests.
- Traffic generation tools: iperf3 (throughput), Scapy scripts (ARP spoof / port scan simulation), a rogue-AP test fixture, and a BLE advertising flooder for controlled anomaly injection.
- Network analyzer (Wireshark) on a separate laptop for out-of-band TLS/encryption verification.
- Battery pack fully charged for endurance/performance test cycles.

11. Responsibilities

Role	Responsibility
Test Lead / QA	Owns the test plan, test case design, defect triage, and final sign-off recommendation.
Backend Developers	Fix defects in capture/analysis/API layers; support unit and integration test automation.
Frontend Developer	Fix dashboard defects; support UI test automation.
Security Tester	Executes Section 5.1 security validation and reports findings.
Project Supervisor / Evaluator	Reviews test summary report and approves Part 1 completion.

12. Staffing and Training Needs

- Team members require working familiarity with Zeek log formats (conn.log, notice.log, weird.log).
- Security tester requires basic familiarity with Nmap, OWASP ZAP, and TLS inspection using Wireshark.
- A short internal knowledge-transfer session is planned before test execution begins to align all testers on the SRS requirement IDs and test case format.

13. Schedule

Milestone	Target Date
Test plan review & sign-off	28 Sept 2026
Test case design complete	T + 3 days
Unit/integration test automation complete	T + 7 days
System & security test execution	T + 10 days
Defect fix & regression cycle	T + 13 days
Final test summary report	T + 15 days

14. Risks and Contingencies

Risk	Impact	Contingency
ESP32 WiFi monitor-mode driver instability	High – blocks WiFi capture test cases	Fall back to host-based WiFi adapter in monitor mode for Part 1 test execution.
Zeek performance bottleneck on Pi hardware under high packet rate	Medium – affects NFR-PERF-1 results	Reduce test load or enable Zeek cluster/lite ruleset for the constrained device.
Limited access to isolated RF test environment	Medium – risk of interference skewing BLE/WiFi results	Schedule tests during low-traffic hours; use a shielded enclosure if available.
Security test tooling unfamiliarity	Low – delays Section 5.1 execution	Pair security tester with a teammate experienced in Nmap/ZAP; consult vendor docs.

15. Approvals

Name/Role	Signature	Date
Test Lead / QA	_____	_____
Project Team Lead	_____	_____
Course Instructor / Evaluator	_____	_____

6. Test Cases and Traceability Matrix

Note on section numbering: per IEEE 829 practice this project groups all detailed test case specifications and the SRS traceability matrix together at the end of the document (referenced from Section 5's approach) so they can be maintained and extended as a single living artifact through Part 2. Ten test cases are specified below, covering functional, non-functional/performance, reliability, and security requirements, each traced to one or more SRS requirement IDs.

6.1 Test Case Specifications

TC-FUNC-01: Start WiFi capture session

Field	Detail
Preconditions	Device powered on; user authenticated
Test Steps	1. Select WiFi interface. 2. Set channel = 6. 3. Click Start Capture.
Expected Result	Capture begins within 2s; live packet counter increments on dashboard.
Traces to Requirement(s)	FR-1, FR-7

TC-FUNC-02: Apply capture filter

Field	Detail
Preconditions	Active or new capture session
Test Steps	1. Enter BPF filter 'tcp port 443'. 2. Start capture. 3. Generate mixed traffic.
Expected Result	Only TCP/443 packets appear in live stats and exported pcap.
Traces to Requirement(s)	FR-4

TC-FUNC-03: BLE advertisement capture

Field	Detail
Preconditions	BLE beacon devices broadcasting nearby
Test Steps	1. Select BLE interface. 2. Set RSSI threshold -70dBm. 3. Start capture.
Expected Result	Only beacons above threshold are logged; others excluded.
Traces to Requirement(s)	FR-3

TC-FUNC-04: Zeek log generation latency

Field	Detail
Preconditions	Active capture with live traffic
Test Steps	1. Start capture. 2. Generate DNS + TCP traffic. 3. Inspect conn.log/dns.log timestamps.
Expected Result	Structured Zeek logs appear within 2 seconds of packet capture.
Traces to Requirement(s)	FR-5

TC-FUNC-05: Export session as PCAP

Field	Detail
Preconditions	Completed capture session
Test Steps	1. Stop capture. 2. Click Export. 3. Open file in Wireshark.

Field	Detail
Expected Result	Valid .pcapng file downloads and opens without corruption.
Traces to Requirement(s)	FR-8

TC-FUNC-06: ARP spoofing detection & alert

Field	Detail
Preconditions	Active capture on Ethernet/WiFi segment
Test Steps	1. Start capture. 2. Run Scapy ARP-spoof script against a test host. 3. Observe dashboard.
Expected Result	Anomaly flagged and alert visible on dashboard within 5 seconds.
Traces to Requirement(s)	FR-10, FR-11

TC-FUNC-07: Alert acknowledgment

Field	Detail
Preconditions	An active unacknowledged alert exists
Test Steps	1. Log in as Field Engineer. 2. Open alert panel. 3. Click Acknowledge.
Expected Result	Alert status changes to Acknowledged; user & timestamp recorded in audit log.
Traces to Requirement(s)	FR-12, SEC-REQ-5

TC-PERF-08: Dashboard update latency under load

Field	Detail
Preconditions	Traffic generator producing 5,000 pkt/s
Test Steps	1. Start capture. 2. Run iperf3 load generator. 3. Measure dashboard stat refresh time.
Expected Result	Dashboard reflects new stats within 2 seconds under the specified load.
Traces to Requirement(s)	NFR-PERF-1

TC-REL-09: Packet loss under sustained load

Field	Detail
Preconditions	Traffic generator producing 1,000 pkt/s for 30 minutes
Test Steps	1. Start capture. 2. Run sustained load for 30 min. 3. Compare sent vs captured packet counts.
Expected Result	Measured packet loss does not exceed 0.1% of packets sent.
Traces to Requirement(s)	NFR-REL-1

TC-SEC-01: Encryption at rest verification

Field	Detail
Preconditions	Completed capture session stored on device
Test Steps	1. Stop capture. 2. Access storage medium directly (e.g., remove SD card, mount read-only). 3. Attempt to read .pcap file and DB contents without the application.
Expected Result	File and database contents are unreadable ciphertext; no plaintext packet data is recoverable.

Field	Detail
Traces to Requirement(s)	SEC-REQ-1, SEC-OBJ-1

TC-SEC-02: Account lockout after failed logins

Field	Detail
Preconditions	Valid user account exists, user logged out
Test Steps	1. Attempt login with wrong password 5 times consecutively. 2. Attempt a 6th login with the correct password.
Expected Result	Account is locked for 5 minutes after the 5th failure; the correct-password attempt is also rejected during lockout, and the lockout event is recorded in the audit log.
Traces to Requirement(s)	SEC-REQ-2, SEC-REQ-6

TC-SEC-03: RBAC boundary enforcement

Field	Detail
Preconditions	Field Engineer and Administrator accounts exist
Test Steps	1. Log in as Field Engineer. 2. Attempt to access the user-management/admin configuration API endpoint directly.
Expected Result	Request is rejected with an authorization error; Field Engineer cannot perform Administrator-only actions.
Traces to Requirement(s)	SEC-REQ-3, SEC-OBJ-2

6.2 SRS – Test Case Traceability Matrix

SRS Requirement ID	Requirement Summary	Covering Test Case(s)
FR-1	WiFi capture in monitor mode	TC-FUNC-01
FR-2	Ethernet capture	TC-PERF-08 (load path exercises Ethernet capture)
FR-3	BLE capture with RSSI filter	TC-FUNC-03
FR-4	Capture filter configuration	TC-FUNC-02
FR-5	Zeek analysis & log generation	TC-FUNC-04
FR-6	Live dashboard statistics	TC-PERF-08
FR-7	Start/stop capture control	TC-FUNC-01
FR-8	PCAP export	TC-FUNC-05
FR-9	Session summary report	Planned for Part 2 regression suite
FR-10	Anomaly detection	TC-FUNC-06
FR-11	Security alert raised	TC-FUNC-06
FR-12	Alert acknowledgment	TC-FUNC-07
FR-13	Storage/retention admin	Planned for Part 2 regression suite
FR-14	Offline local-AP connectivity	Verified via environment setup in Section 10

SRS Requirement ID	Requirement Summary	Covering Test Case(s)
FR-15	Concurrent multi-interface capture	TC-PERF-08 (WiFi+Ethernet concurrently active)
NFR-PERF-1	Dashboard update latency ≤ 2s @ 5,000 pkt/s	TC-PERF-08
NFR-REL-1	Packet loss ≤ 0.1%	TC-REL-09
SEC-REQ-1 / SEC-OBJ-1	Encryption at rest	TC-SEC-01
SEC-REQ-2 / SEC-REQ-6	Authentication & lockout policy	TC-SEC-02
SEC-REQ-3 / SEC-OBJ-2	RBAC enforcement	TC-SEC-03
SEC-REQ-5	Tamper-evident audit log	TC-FUNC-07, TC-SEC-02

This matrix demonstrates bidirectional traceability between the SRS (PPS-SRS-1.0) and this Test Plan: every functional, performance, reliability, and security requirement introduced in the SRS maps to at least one test case defined in Section 6.1, and every test case above traces back to a specific requirement ID.